Chaudhary Varun

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.97
Shree Sarvajanik Vardhman Vidyalaya	Class XII (Gujarat Secondary & Higher Secondary Education Board)	2016	74.00%
Shree Sarvajanik Vardhman Vidyalaya	Class X (Gujarat Secondary & Higher Secondary Education Board)	2014	79.00%

PROJECTS

Two-Way Fluid-Structure Interaction (FSI) Analysis of a Flexible Wing

M.Tech Thesis

- Developing a two-way partitioned FSI framework coupling **OpenFOAM (CFD) and CalculiX (FEA)** through **preCICE** to capture dynamic aeroelastic deformation of flapping-wing mechanisms.
- Investigating wing flexibility configurations and coupling convergence to evaluate gains in net lift and thrust; automating transient force analysis with Python scripts.

Material Characterisation and Test-CAE Correlation of Windshield PVB Layer

M.Tech Project

- Performed experimental modal testing and **Abaqus-based numerical validation** of laminated windshields to evaluate frequency-dependent damping and vibration sensitivity in the low-frequency regime (<200 Hz).
- **Characterized** the viscoelastic behaviour of **PVB interlayers** using **Prony-series parameters**, achieving high-fidelity correlation between experimental modal analysis and Abaqus dynamic simulations.

Quasi-Static Crushing Simulation of CFRP Composite Cylinders

MTech Project

- Modelled axial crushing of CFRP composite cylinders in Abaqus/Explicit using the Hashin damage criteria to evaluate progressive failure mechanisms, Specific Energy Absorption (SEA), and Peak Crush Force (PCF).

Numerical Analysis of a Vertical Axis Wind Turbine (Ansys Fluent)

BTech Project

- Performed CFD-based numerical analysis of a **Complex vertical-axis wind turbine (VAWT)** in Ansys Fluent to evaluate aerodynamic performance under complex flow conditions.

TECHNICAL SKILLS

Modeling & Numerical Methods: Finite Element Method (**FEM**), Finite Volume Method (**FVM**), Fluid - Structure Interaction (**FSI**), Multi-physics modeling, Partitioned coupling, Transient simulation.

Simulation & Software: Abaqus, Ansys Mechanical and Fluent, Calculix, preCICE

CAD & Post-Processing: Solidworks, Siemens NX, Techplot

Programming & Environment: Python, MATLAB, Linux, Windows, HPC job submission (SLURM), Automation, Solver development

Simulation & Software: Abaqus, Ansys Mechanical and Fluent, OpenFOAM, CalculiX, preCICE, Ansys ICEPACK

FEA Customisation: Abaqus UMAT/UHYPER development for custom material models using Fortran

RELEVANT COURSEWORK

FEM | Analysis & Design of Composite Structure | AFEM | Fracture Mechanics | Vibration and Dynamics | CFD | ACFD |

PROFESSIONAL EXPERIENCE

Mechanical Technical Specialist | Indian Railways

Gorakhpur, India

- Created and modified mechanical drawings in Siemens NX in the Jigs & Fixtures Shop, supporting fabrication, maintenance, and shop-floor distribution of custom tooling drawings.
- Performed fault diagnosis, part replacement, and functional testing in the Air Brake and Bogie Shop using air brake testing rigs, ensuring reliable brake performance in passenger and freight vehicles.

CERTIFICATION AND WORKSHOPS

- Introduction to Fluid-Structure Interaction using OpenFOAM - NPTEL+ Workshop.
- Mastering Ansys CFD - Udemy.
- MATLAB Onramp - MathWorks.
- Fundamentals of Fluid-Solid Interaction - École Polytechnique (Coursera).

AWARDS

- Inter-IIT Basketball Championship - IIT Madras,
- Inter-IIT Basketball Championship - IIT Kanpur, Dec 2024
- Runner-up, All India Inter-University Basketball Championship - Association of Indian Universities.

POSITION OF RESPONSIBILITY

MAE M.Tech Head Placement Coordinator, IITH

- Coordinated end-to-end campus recruitment as the primary liaison among students, faculty, alumni and HR teams, strengthening alumni-industry connections to attract new recruiters and expand placement opportunities.

Inter-IIT Sports Meet -- IIT Madras | Basketball Team Captain

- Led the IIT Hyderabad Basketball Team as Captain to the **Semi-Finals of the Inter-IIT Sports Meet at IIT Madras**, guiding team strategy, coordination, and on-court execution throughout the tournament.

LANGUAGES

- **Fluent:** English, Hindi, Gujarati
- **Conversational:** Tamil

EXTRACURRICULAR

Basketball, Volleyball, Fitness Training, and Travelling